



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2023
ASSESSMENT TASK 1

Mathematics Extension 2

General Instructions

- **Working time – 60 minutes**
Reading time – 5 minutes
- Write on both sides of the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators only
- All necessary working should be shown in every question
- Each new question is to be started on a **new booklet**.
- Attempt all questions.

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Umakanthan
- ☐ Ms Lee

Student Number:

(To be used by the exam markers only.)

Question	1	2	3	Total	Percent
Mark	$\overline{14}$	$\overline{11}$	$\overline{12}$	$\overline{37}$	$\overline{100}$

Question 1 (14 Marks)

(a) Let $z = 4 + 2i$. Find:

(i) $z\bar{z}$

1

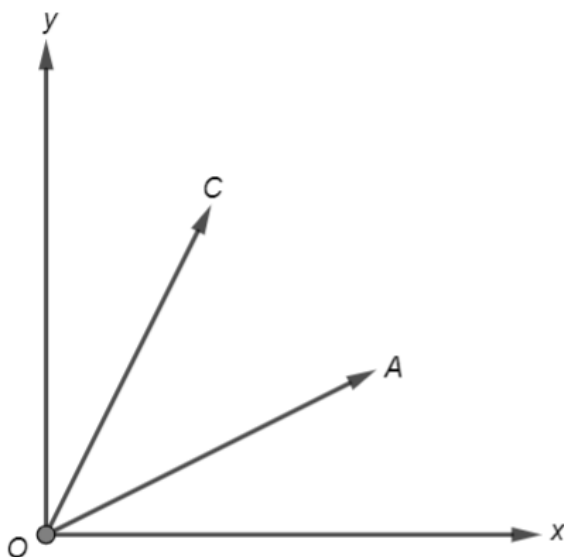
(ii) $z - \bar{z}$

1

(b) Write $\frac{3+3i}{2+i}$ in the form $x + iy$ where x and y are real numbers

1

(c) If O is the origin and $\overrightarrow{OA} = 2 + i$ and $\overrightarrow{OC} = 1 + 2i$



(i) Find B such that $OABC$ is a rhombus

1

(ii) Show that $\tan \angle AOB = \frac{1}{3}$

2

(iii) Hence show that $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{2} = \frac{\pi}{4}$

2

(d) Given that $\frac{z^2}{z-1}$ is a real number, sketch the locus of z on the Argand diagram.

3

(e) Solve the equation $z + 1 + 8i = |z|(1 + i)$ where $z \in \mathbb{C}$

3

Question 2 (11 Marks)

- (a) $y = f(x)$ is a function. Consider the following 2 statements

$p: f''(a) = 0$

q : There is a point of inflection at $(a, f(a))$

Which of the following statements are true

2

1. p is a necessary condition for q
2. q is a necessary condition for p
3. p is a sufficient condition for q
4. q is a sufficient condition for p

- (b) Consider the following statement:

If I study Extension 2 Mathematics, then I am super-cool.

Write down the:

(i) Converse

1

(ii) Negation

1

(iii) Contrapositive

1

- (c) Prove that $a^2 - 1$ is divisible by 3 if and only if a is not divisible by 3 where $a \in \mathbb{Z}$

3

- (d) (i) Prove that if $a \mid b$ and $a \mid c$ then $a \mid b - c$ where a, b and c are integers

1

- (ii) Hence or otherwise prove that $n! + 1$ and $(n + 1)! + 1$ are relatively prime where n is a positive integer.

2

* $a \mid b$ means ' a divides b ' so a is a factor of b where a and b are both integers

** two numbers are relatively prime if they have no common integer factors other than 1

Question 3 (12 Marks)

(a) If w is a non-real root of the equation $z^3 = 1$

(i) Show that $1 + w + w^2 = 0$

1

(ii) Find the value of $(1 - w)(1 - w^2)(1 - w^4)(1 - w^8)$

2

(b) (i) By considering the expansion of $(\cos \theta + i \sin \theta)^7$ or otherwise:

Write $\frac{\sin 7\theta}{\sin \theta}$ in terms of $\sin \theta$.

2

(ii) Hence or otherwise find the roots of the equation $64x^3 - 112x^2 + 56x - 7 = 0$

2

(iii) Hence find $\operatorname{cosec}^2 \frac{\pi}{7} + \operatorname{cosec}^2 \frac{2\pi}{7} + \operatorname{cosec}^2 \frac{3\pi}{7}$

2

(c) Show that if m and n are positive integers there are no complex numbers z , such that:

$$z^n = 1 + i \quad \text{and}$$

$$z^m = 2 - i$$

3

Question 1

$$\begin{aligned} \text{(a) (i)} \quad z \bar{z} &= (4+2i)(4-2i) \\ &= 4^2 - (2i)^2 \\ &= 16 + 4 \\ &= 20 \end{aligned}$$

$$\begin{aligned} \text{(a) (ii)} \quad z - \bar{z} &= (4+2i) - (4-2i) \\ &= \cancel{4} + 2i - \cancel{4} + 2i \\ &= 4i \end{aligned}$$

$$\text{(b)} \quad \frac{3+3i}{2+i} = \frac{3+3i}{2+i} \times \frac{(2-i)}{(2-i)}$$

$$= \frac{(3+3i)(2-i)}{2^2 - i^2}$$

$$= \frac{6 - 3i + 6i + 3}{4 + 1}$$

$$= \frac{9 + 3i}{5}$$

$$= \frac{9}{5} + \frac{3}{5}i$$

$$\begin{aligned}
 \text{(c) (i)} \quad \vec{OB} &= \vec{OA} + \vec{AB} \\
 &= (2+i) + (1+2i) \quad (\text{as } \vec{AB} = \vec{OC}) \\
 &= 3+3i
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \tan \angle AOB &= \tan (\angle xOB - \angle xOA) \\
 &= \tan (\text{Arg } \vec{OB} - \text{Arg } \vec{OA}) \\
 &= \tan \left(\text{Arg} \left(\frac{\vec{OB}}{\vec{OA}} \right) \right) \\
 &= \tan \left(\text{Arg} \left(\frac{3+3i}{2+i} \right) \right) \\
 &= \tan \left(\text{Arg} \left(\frac{9}{5} + \frac{3}{5}i \right) \right) \quad (\text{from b}) \\
 &= \frac{3/5}{9/5} \quad \left(\text{as } \tan \text{Arg}(z) = \frac{\text{Im}(z)}{\text{Re}(z)} \right) \\
 &= \frac{3}{9} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$\text{(iii)} \quad \angle AOB = \tan^{-1} \left(\frac{1}{3} \right) \quad (\text{from (ii)})$$

$$\angle xOA = \tan^{-1} \left(\frac{1}{2} \right)$$

$$\angle xOB = \tan^{-1} \left(\frac{3}{3} \right)$$

$$= \tan^{-1}(1)$$

$$= \pi/4$$

$$\angle AOB + \angle xOA = \angle xOB \quad \checkmark$$

$$\tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{4}$$

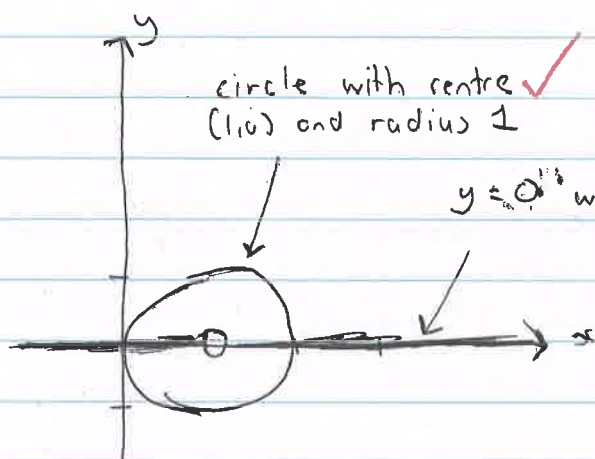
$$(d) \frac{z^2}{z-1} = \frac{(x+iy)^2}{x+iy-1} \quad (\text{letting } z = x+iy)$$

$$= \frac{x^2 - y^2 + 2xyi}{(x-1) + iy} \times \frac{(x-1) - iy}{(x-1) - iy}$$

$$\text{numerator} = (x^2 - y^2 + 2xyi)(x-1-iy)$$

$$\begin{aligned} \Im(\text{numerator}) &= 2x^2y - 2xy - x^2y + y^3 \\ &= y^3 + x^2y - 2xy \\ &= y(y^2 + x^2 - 2x) \end{aligned} \quad \checkmark$$

$$\begin{aligned} y &= 0 \quad \text{or} \quad y^2 + x^2 - 2x = 0 \\ y^2 + x^2 - 2x + 1 &= 1 \\ y^2 + (x-1)^2 &= 1 \end{aligned}$$



circle with centre $(1, 0)$ and radius 1 $\checkmark$

$y=0$ with open circle at $(1, 0)$ $\checkmark$

$$(e) \quad z + 1 + 8i = |z|(1+i)$$

$$z + 1 + 8i = |z| + |z|i$$

$$z = |z| + |z|i - 1 - 8i$$

$$z = (|z| - 1) + (|z| - 8)i$$

$$|z|^2 = (|z| - 1)^2 + (|z| - 8)^2$$

$$|z|^2 = |z|^2 - 2|z| + 1 + |z|^2 - 16|z| + 64$$

$$|z|^2 - 18|z| + 65 = 0$$

$$(|z| - 5)(|z| - 13) = 0$$

$$|z| = 5 \text{ or } |z| = 13$$

$$z = 4 - 3i \text{ or } z = 12 + 5i$$

Question 2

(a) 1 & 4



(b) (i) If I am super-cool then I study Extension 2 Mathematics



(b) (ii) I study Extension 2 Mathematics and I am not super-cool.



(b) (iii) If I am not super-cool then I do not study Extension 2 Mathematics.



(c) $a^2 - 1$ is divisible by 3 $\Rightarrow$ a is not divisible by 3
take the contrapositive:

a is divisible by 3 $\Rightarrow a^2 - 1$ is not divisible by 3

let $a = 3k$, $k \in \mathbb{Z}$

$$\therefore a^2 - 1 = (3k)^2 - 1$$

$$= 9k^2 - 1$$

$$= 3(3k^2) - 1$$

not divisible by 3

$\therefore a^2 - 1$ is divisible by 3 $\Rightarrow a$ is not divisible by 3 is true by contrapositive proof



a is not divisible by 3 $\Rightarrow a^2 - 1$ is divisible by 3

let $a = 3k \pm 1$, $k \in \mathbb{Z}$

$$a^2 - 1 = (3k \pm 1)^2 - 1$$

$$= 9k^2 \pm 6k + 1 - 1$$

$$= 9k^2 \pm 6k$$

$$= 3(3k^2 \pm 2)$$

divisible by 3

$\therefore a$ is not divisible by 3 $\Rightarrow a^2 - 1$ is divisible by 3



attempting both ways

$\therefore a^2 - 1$ is divisible by 3 $\Leftrightarrow a$ is not divisible by 3.

$$(d)(i) a|b \Rightarrow b = pa \quad p \in \mathbb{Z} \quad p \neq 0$$

$$a|c \Rightarrow c = qa \quad q \in \mathbb{Z} \quad q \neq 0$$

$$\begin{aligned} b - c &= pa - qa \\ &= a(p - q) \end{aligned}$$



$$\therefore a | b - c$$

(d)(ii) assume the negation

i.e. assume $n! + 1$ and $(n+1)! + 1$ have a common factor.

by (i) this common factor must be shared by:

$$\begin{aligned} (n+1)! + 1 - (n! + 1) &= (n+1)n! + 1 - n! - 1 \\ &= n!(n+1-1) \\ &= n \times n! \end{aligned}$$



but this is a contradiction as $n! + 1$ and $n \times n!$ clearly have no common factors (other than 1)

$n! + 1$ and $(n+1)! + 1$ $\nmid$ also have no common factors other than 1 and hence are relatively prime.

Question 3

(a) (i) $z^3 = 1$

$\therefore w^3 = 1$ (w is a solution to the equation)

$$w^3 - 1 = 0$$

$$(w-1)(w^2+w+1) = 0$$

$$\therefore w-1 = 0 \quad \text{or} \quad w^2+w+1 = 0$$

but w is non real $\therefore w-1 \neq 0$

$$\therefore w^2 + w + 1 = 0 \quad \checkmark$$

(a) (ii) $(1-w)(1-w^2)(1-w^4)(1-w^8)$

$$\begin{aligned} &= (1-w)(1-w^2)(1-w)(1-w^2) \quad \left(\begin{array}{l} w^4 = w \\ w^8 = w^2 \end{array} \right) \quad \checkmark \\ &= (1-w)^2 (1-w^2)^2 \\ &= (1-2w+w^2)(1-2w^2+w^4) \\ &= (1-2w^2+w^2)(1-2w^2+w) \\ &= 1-2w^2+w-2w+4w^3-2w^2+w^2-2w^4+w^3 \\ &= \underline{1} - 2w^2 + \underline{w} - \underline{2w} + \underline{4} - 2w^2 + w^2 - \underline{2w} + \underline{1} \\ &= 6 - 3w - 3w^2 \\ &= 6 - 3(w+w^2) \\ &= 6 - 3(-1) \quad \text{as } (w^2+w = -1 \text{ from (i)}) \\ &= 9 \quad \checkmark \end{aligned}$$

$$(b)(i) (\cos \theta + i \sin \theta)^7 = \cos 7\theta + i \sin 7\theta \quad (\text{De Moivre's Theorem})$$

$$(\cos \theta + i \sin \theta)^7 = \binom{7}{0} \cos^7 \theta + \binom{7}{1} \cos^6 \theta i \sin \theta - \binom{7}{2} \cos^5 \theta \sin^2 \theta + \binom{7}{3} \cos^4 \theta i \sin^3 \theta + \binom{7}{4} \cos^3 \theta \sin^4 \theta + \binom{7}{5} \cos^2 \theta i \sin^5 \theta - \binom{7}{6} \cos \theta \sin^6 \theta + \binom{7}{7} i \sin^7 \theta$$

equate imaginary parts

$$\sin 7\theta = 7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta \quad \checkmark$$

$$\therefore \frac{\sin 7\theta}{\sin \theta} = 7 \cos^6 \theta - 35 \cos^4 \theta \sin^2 \theta + 21 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$$

$$= 7(1 - \sin^2 \theta)^3 - 35 \sin^2 \theta (1 - \sin^2 \theta)^2 + 21 \sin^4 \theta (1 - \sin^2 \theta) - \sin^6 \theta$$

$$= 7(1 - 3\sin^2 \theta + 3\sin^4 \theta + \sin^6 \theta) - 35 \sin^2 \theta (1 - 2\sin^2 \theta + \sin^4 \theta) + 21 \sin^4 \theta - 21 \sin^6 \theta - \sin^6 \theta$$

$$= 7 - 21 \sin^2 \theta + 21 \sin^4 \theta + 7 \sin^6 \theta - 35 \sin^2 \theta + 70 \sin^4 \theta - 35 \sin^6 \theta + 21 \sin^4 \theta - 21 \sin^6 \theta - \sin^6 \theta$$

$$= 7 - 56 \sin^2 \theta + 112 \sin^4 \theta - 64 \sin^6 \theta \quad \checkmark$$

$$(b)(ii) \text{ Letting } \sin 7\theta = 0 \text{ and } x = \sin^2 \theta \text{ where } \sin \theta \neq 0 \quad \checkmark$$

$$0 = 7 - 56x + 112x^2 - 64x^3$$

$$\therefore 64x^3 - 112x^2 + 56x - 7 = 0$$

$$\text{as } \sin 7\theta = 0$$

$$7\theta = k\pi \quad -7 < k \leq 7$$

$$\theta = k\pi/7 \quad (k \neq 0 \text{ as } \sin \theta \neq 0)$$

$$\therefore \theta = \pm \pi/7, \pm 2\pi/7, \pm 3\pi/7 \quad (\text{other values of } k \text{ will give duplicates})$$

$$\therefore x = \sin^2 \frac{\pi}{7}, \sin^2 \frac{2\pi}{7} \text{ or } \sin^2 \frac{3\pi}{7} \quad (\text{negative values of } \sin \text{ become positive}) \quad \checkmark$$

$$(b)(iii) \operatorname{cosec}^2 \frac{\pi}{7} + \operatorname{cosec}^2 \frac{2\pi}{7} + \operatorname{cosec}^2 \frac{3\pi}{7}$$

$$= \frac{1}{\sin^2 \pi/7} + \frac{1}{\sin^2 2\pi/7} + \frac{1}{\sin^2 3\pi/7}$$

$$= \frac{\sin^2 \pi/7 \sin^2 2\pi/7 + \sin^2 \pi/7 \sin^2 3\pi/7 + \sin^2 2\pi/7 \sin^2 3\pi/7}{\sin^2 \pi/7 \times \sin^2 2\pi/7 \times \sin^2 3\pi/7} \quad \checkmark$$

$$= \frac{56/64}{-(-7)/64}$$

$$= 8 \quad \checkmark$$

(c) Proof by contradiction:

assume $\exists n \text{ \& } m$ such that $m, n \in \mathbb{Z}$

$$\text{let } z = r \operatorname{cis} \theta$$

$$z^n = r^n \operatorname{cis} n\theta$$

$$z^m = r^m \operatorname{cis} m\theta \quad \checkmark$$

$$r^n \operatorname{cis} n\theta = 1 + i$$

$$r^n = \sqrt{2}$$

$$r^n = 2^{1/2}$$

$$r = 2^{1/2n}$$

$$r^m \operatorname{cis} m\theta = 2 - i$$

$$r^m = \sqrt{5}$$

$$r^m = 5^{1/2}$$

$$r = 5^{1/2m} \quad \checkmark$$

$$\therefore 2^{1/2n} = 5^{1/2m} \quad \leftarrow \text{both sides to power of } 2mn$$

$$\therefore 2^m = 5^n$$

contradiction $\checkmark$

$\therefore$ no such integers exist.